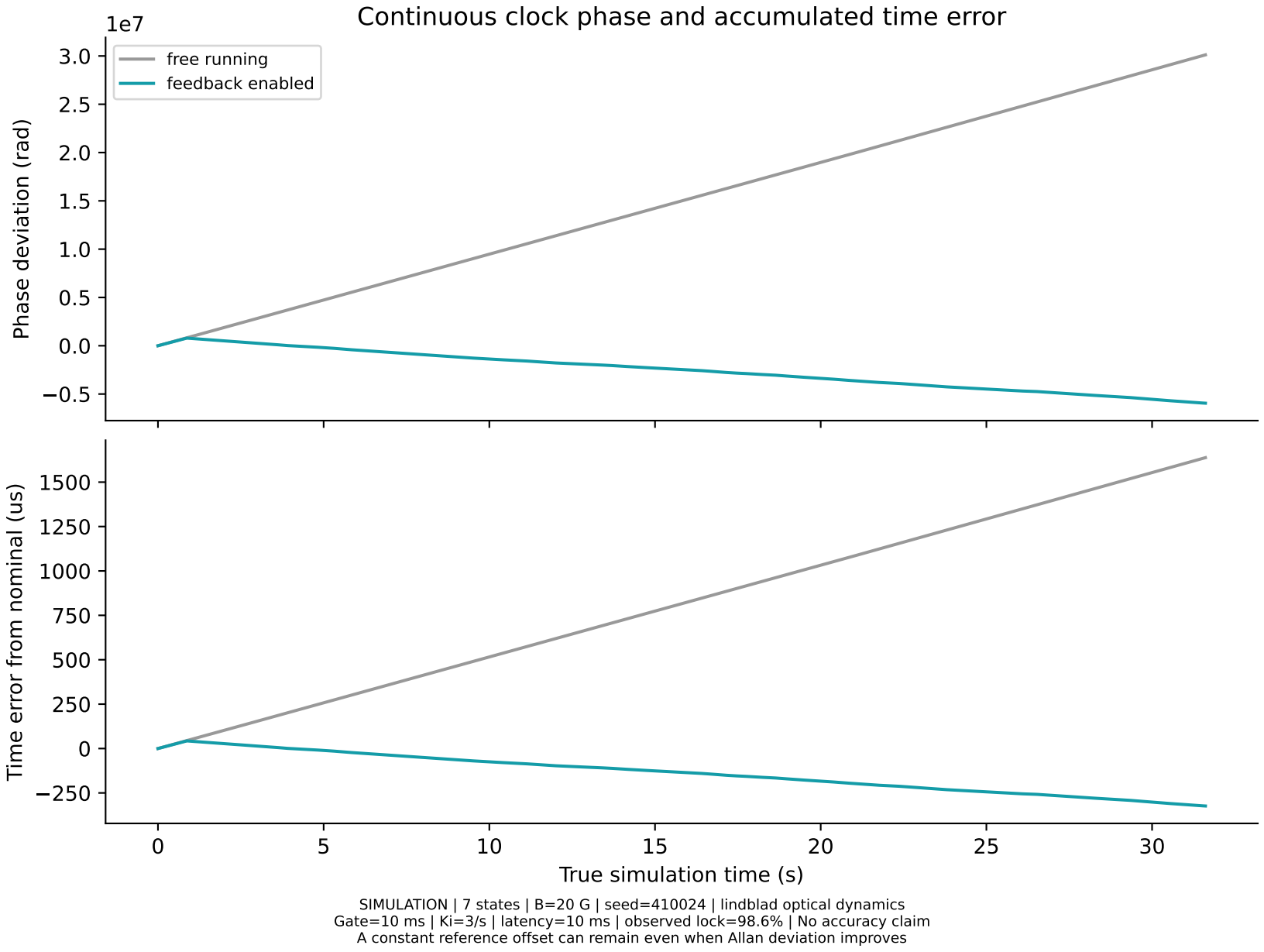
Continuous clock phase and accumulated time error
1e7
free running
feedback enabled
Phase deviation (rad)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
Time error from nominal (us)
1500
1250
1000
750
500
250
0
−250
0
5
10
15
20
25
30
True simulation time (s)
SIMULATION | 7 states | B=20 G | seed=410024 | lindblad optical dynamics
Gate=10 ms | Ki=3/s | latency=10 ms | observed lock=98.6% | No accuracy claim
A constant reference offset can remain even when Allan deviation improves